

Lecture 6: **Monte Carlo methods**

September 14, 2026

in-class: escape the local minima!

$$f(x) = \sin(5x) + 0.1x^2$$

try running BFGS from 10 evenly spaced starting points across

$x \in [-10, 10]$ and plot the result

```
scipy.optimize.minimize(method='BFGS')
```

Frequentist vs Bayesian: maximum likelihood estimation (MLE) and maximum a posteriori (MAP)

```
In [22]: # again, maximize likelihood = minimize negative likelihood
def neg_log_posterior(theta, F, e):
    return -log_posterior(theta, F, e)

theta_guess = [900, 20] # initial guess
theta_est = sp.optimize.fmin(neg_log_posterior, theta_guess, args=(F, e))
print("""
    Maximum A Posteriori (MAP) estimate for {0} data points:
        mu={theta[0]:.0f}, sigma={theta[1]:.0f}
    """).format(N, theta=theta_est))
```

```
Optimization terminated successfully.
    Current function value: 482.290553
    Iterations: 53
    Function evaluations: 101
```

```
Maximum A Posteriori (MAP) estimate for 100 data points:
    mu=1002, sigma=11
```

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we want more than MAP — mean, variance, confidence interval, and other **uncertainty quantification** of the posterior

```
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Maximum A Posteriori (MAP) estimate for 100 data points:
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uncertainty quantification requires evaluating the expectation value of some f over the posterior $p(\theta | X)$

$$\mathbb{E}_{p(\theta|X)}(f) = \int f(\theta) p(\theta | X) \mathrm{d}\theta$$

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$$\mathbb{E}_{p(\theta|X)}(f) = \int f(\theta) p(\theta | X) d\theta$$

mean

$$\mathbb{E}_{p(\theta|X)}(\theta) = \int \theta p(\theta | X) d\theta$$

confidence
interval

$$\mathbb{E}_{p(\theta|X)}(l < \theta < u) = \int_l^u p(\theta | X) d\theta$$

marginalization is also an important feature of Bayesian inference

$$\theta = \{\theta', z\}$$

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parameters of interest “nuisance” or “latent” parameters
e.g. instrument calibration

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parameters of interest “nuisance” or “latent” parameters

$$p(\theta' | X) = \int p(\theta', z | X) dz$$

also, posterior predictive distribution

$$p(X_{\text{new}} | X) = \int p(X_{\text{new}} | \theta) p(\theta | X) d\theta$$

probability distribution of future data, X_{new} , given observed data X

“what do I expect to observe next?”

analytic or numerical integration quickly becomes
intractable for high dimensions and complex likelihoods

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we can instead use **Monte Carlo** methods



Casino de Monte Carlo, Monaco

Monte Carlo methods were first developed for statistical methods in Los Alamos

“I wondered whether the probability of success could be estimated by playing many games...”

Stanislaw Ulam, *Adventures of a Mathematician*



Casino de Monte Carlo, Monaco

Monte Carlo methods can be used to expectation value of any f of a random variable x drawn from some distribution $p(x)$

$$\mathbb{E}_{p(x)}(f) = \int f(x) p(x) \, dx$$

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$$\mathbb{E}_{p(x)}(f) = \int f(x) p(x) \, dx \quad \approx \quad \frac{1}{S} \sum_{i=1}^S f(x_i)$$

$x_1, x_2, \dots, x_S$ are S samples drawn from $p(x)$

in-class: estimate π using Monte Carlo methods

hint: $f(x, y) = \mathbf{1}(x^2 + y^2 \leq r^2)$ and $p(x) = p(y) = \mathcal{U}(-r, r)$

Monte Carlo methods can be used to expectation value of any f of a random **variable** x **drawn from some distribution** $p(x)$

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how can we sample the posterior?

sampling standard distributions: *sample a simple distribution*
then apply a transformation

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cumulative distribution function (CDF)

$$\Phi(\theta) = p(\theta' < \theta)$$

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$$u \sim \mathcal{U}(0,1) \quad \rightarrow \quad x = \Phi^{-1}(u) \sim p(x)$$

exponential distribution ($\lambda e^{-\lambda x}$):

$$u = 1 - e^{-\lambda x}$$

$$x = -\ln(1 - u)/\lambda$$

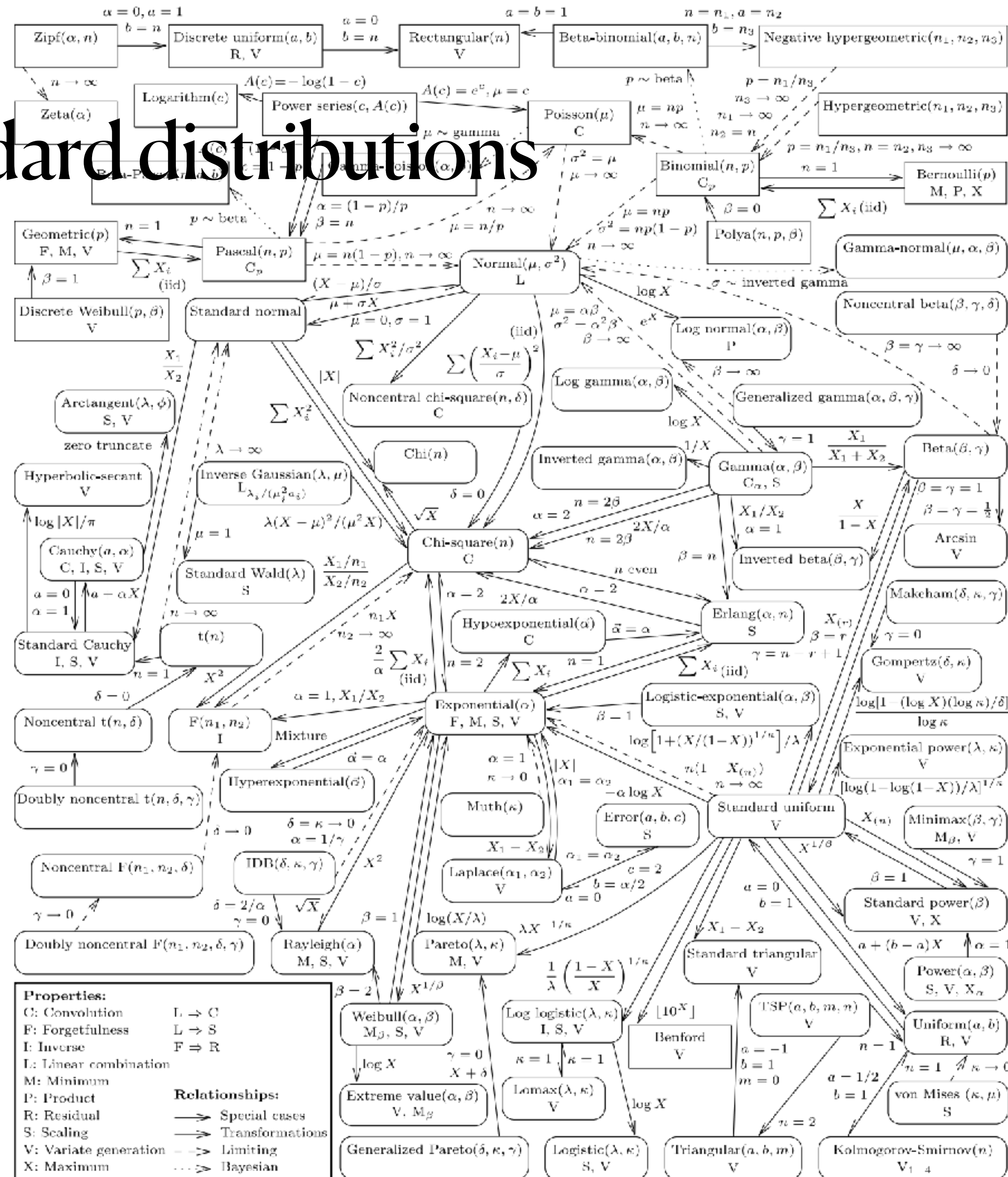
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χ^2 distribution:

$$x = -2 \ln(u)$$

sampling standard distributions



how can we sample **non-standard** probability distributions?

rejection sampling of $p(x)$

rejection sampling of $p(x)$

create a **proposal distribution** $q(x)$ that satisfies $M q(x) \geq \tilde{p}(x)$

rejection sampling of $p(x)$

some constant

create a **proposal distribution** $q(x)$ that satisfies $M q(x) \geq \tilde{p}(x)$

unnormalized $p(x)$

rejection sampling of $p(x)$

create a **proposal distribution** $q(x)$ that satisfies $M q(x) \geq \tilde{p}(x)$

sample $x' \sim q(x)$ and $u \sim U(0,1)$

rejection sampling of $p(x)$

create a **proposal distribution** $q(x)$ that satisfies $M q(x) \geq \tilde{p}(x)$

sample $x' \sim q(x)$ and $u \sim U(0,1)$

if $u > \frac{\tilde{p}(x')}{Mq(x')}$ **reject** x'

rejection sampling of $p(x)$

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if $u > \frac{\tilde{p}(x')}{Mq(x')}$ **reject** x' ; otherwise, **accept**

jupyter notebook on rejection sampling

rejection sampling for Bayesian inference

$$\tilde{p}(x) = p(X \mid \theta) p(\theta) = p(\theta \mid X) / p(X)$$

rejection sampling for Bayesian inference

$$\tilde{p}(x) = p(X | \theta) p(\theta) = p(\theta | X) / p(X)$$

$$q = p(\theta) \text{ with } M = p(X | \theta_{\text{MLE}})$$

prior likelihood at MLE

rejection sampling for Bayesian inference

$$\tilde{p}(x) = p(X \mid \theta) p(\theta) = p(\theta \mid X) / p(X)$$

$$q = p(\theta) \text{ with } M = p(X \mid \theta_{\text{MLE}})$$

$$\text{sample } \theta' \sim p(\theta) \text{ and } u \sim U(0,1)$$

rejection sampling for Bayesian inference

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$$q = p(\theta) \text{ with } M = p(X \mid \theta_{\text{MLE}})$$

sample $\theta' \sim p(\theta)$ and $u \sim U(0,1)$

$$\text{if } u > \frac{\tilde{p}}{Mq} = \frac{p(X \mid \theta)}{p(X \mid \theta_{\text{MLE}})} \text{ reject; else accept}$$

rejection sampling for Bayesian inference

even with a clever choice of M q , a lot of samples are rejected

inefficient, especially in higher dimensions

importance sampling offers a more efficient sampling method

importance sampling

choose a proposal distribution $q(x)$ that is easy to sample

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importance sampling

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$$\mathbb{E}_{p(x)}(f) = \int f(x) p(x) \, dx = \int f(x) \frac{p(x)}{q(x)} q(x) \, dx$$

importance sampling

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$x_i \sim q(x)$ and $w_i = \frac{p(x_i)}{q(x_i)}$ are the **importance weights**

importance sampling

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$x_i \sim q(x)$ and $w_i = \frac{p(x_i)}{q(x_i)}$ are the **importance weights**

unlike rejection sampling, all samples are used

importance sampling for Bayesian inference

$$q = p(\theta) \quad \text{prior}$$

$$w_i = \frac{p(X | \theta) p(\theta)}{p(\theta)} = p(X | \theta) \quad \text{likelihood}$$

importance sampling

If q is a poor match for p only a small fraction of samples will contribute to estimating $\mathbb{E}_p(f)$

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importance weights $w = \frac{p}{q}$ can vary by large factors

nevertheless, some **applications of importance sampling**
in astronomy

reweighting between models: *e.g.* $\Lambda\text{CDM} \rightarrow \nu\Lambda\text{CDM}$

combined multi-probe analyses
(*e.g.* joint CMB + galaxy clustering posterior)

updating priors of existing posterior samples

in-class: sample a Kroupa IMF using rejection sampling

reweighting between models: *e.g.* $\Lambda\text{CDM} \rightarrow \nu\Lambda\text{CDM}$

$$\xi(m) \propto \begin{cases} (m/m_{\text{break}})^{-\alpha_1}, & m_{\text{min}} \leq m < m_{\text{break}} \\ (m/m_{\text{break}})^{-\alpha_2}, & m_{\text{break}} \leq m \leq m_{\text{max}} \end{cases}$$

$$\alpha_1 = 1.3, \alpha_2 = 2.3, m_{\text{break}} = 0.5 M_{\odot} \text{ sample over } [0.08, 100] M_{\odot}$$

next class: **Markov Chain Monte Carlo (MCMC)**